

Continuous Integration and Delivery (CI/CD) with GitHub Actions

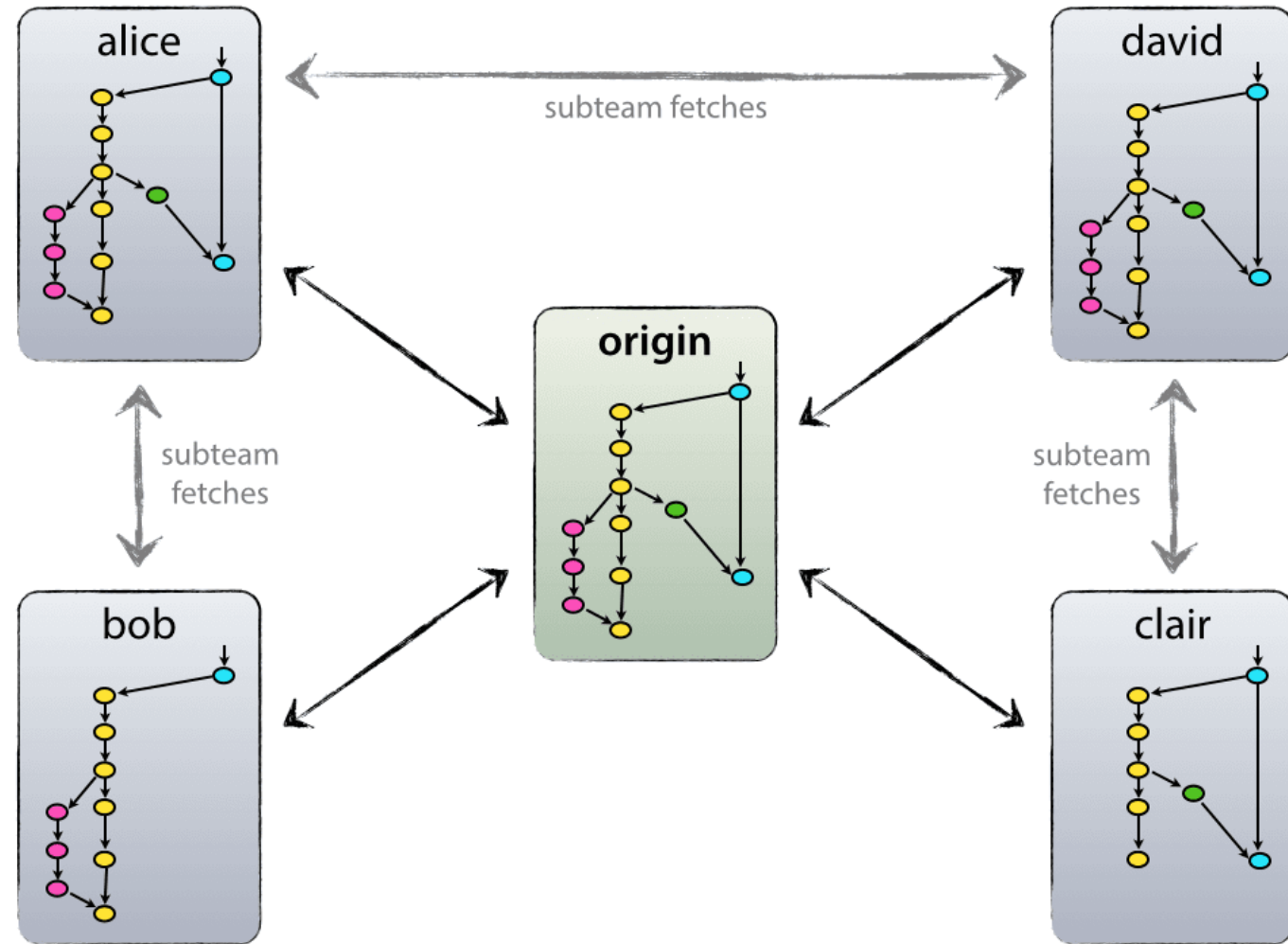
Federico A. Galatolo

Git

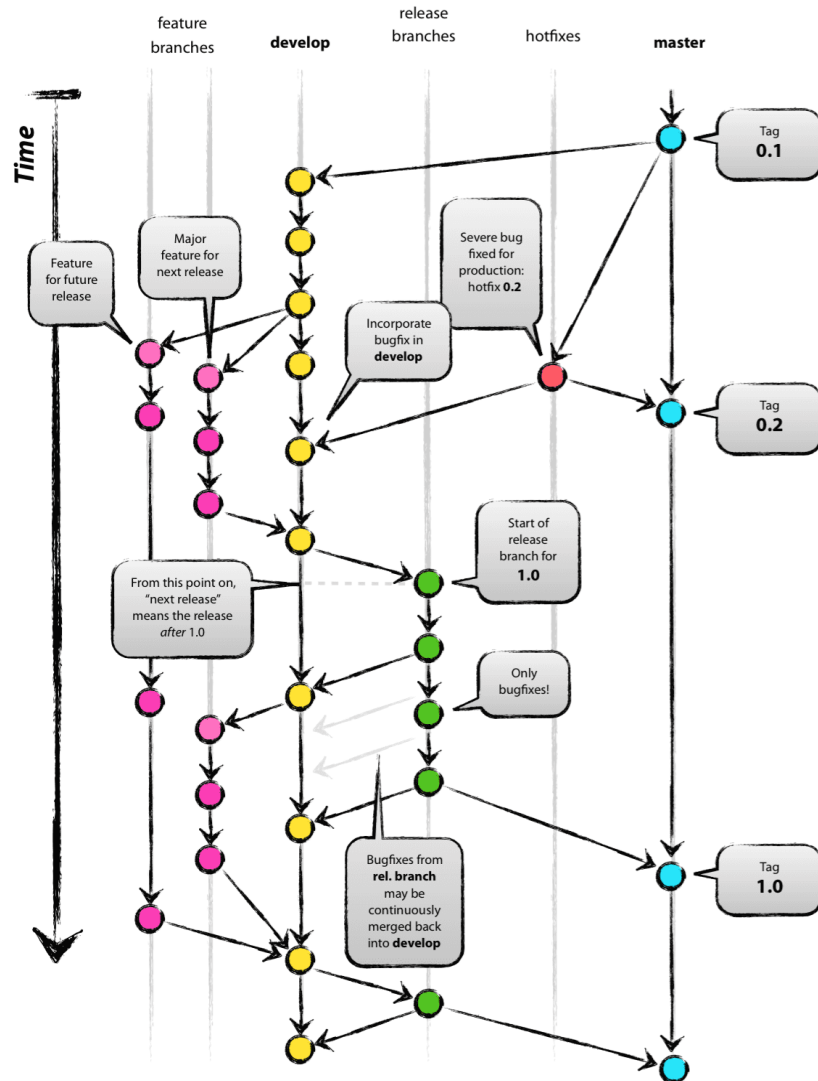
- Widespread Version Control Software (VCS)
- Developed by Linus Torvalds and co. as the Linux Kernel VCS
- It is a **distributed** VCS, even if most of times it is used in a centralized way



Git Workflow



Git Branches and Tags



- This is known as “A successful Git branching model”
- Can be quite challenging to manage
- Only **stable** versions on the master branch
- Hotfixes are **solved** in the hotfixes branch and directly merged on master
- Only **working** versions on the develop branch
- Features are developoed in features branch (one branch per feature)

Continuous Integration and Delivery

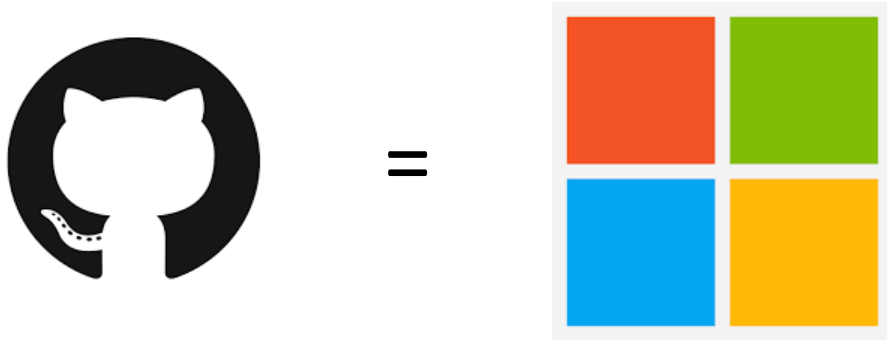
- Run sequence of actions in response to repository events
 - commits
 - merges
 - pull requests
 - issue creation
 - etc..
- Continuous Integration
 - Run tests as soon as code is merged on develop
 - Reject failing commits
 - Update code metrics (coverage, inheritance depth, complexity, etc...)
 - Automatically merge code when needed (for example automerge hotfixes in develop)
 - etc..
- Continuous Delivery
 - Run tests, build and ship software as soon as code is merged on master
 - Run A/B delivery
 - Run Incremental Delivery
 - etc...

CI/CD Providers Comparison

Provider	Open Source	Self-Hostable	Free CI/CD minutes month	Additional minutes prices
GitHub	No	No	2.000	0.008 \$/minute
Bitbucket	No	No	50	0.01 \$/minute
GitLab (hosted)	Yes	Yes	400	0.01 \$/minute
GitLab (self-hosted)	Yes	Yes	unlimited*	free*

Provider	Use Case
GitHub	small-medium open source projects. small business
Bitbucket	business that uses atlassian's tools (jira, confluence, etc...)
GitLab (hosted)	small open source projects. small-medium business
GitLab (self-hosted)	medium-big open source projects. medium-big business

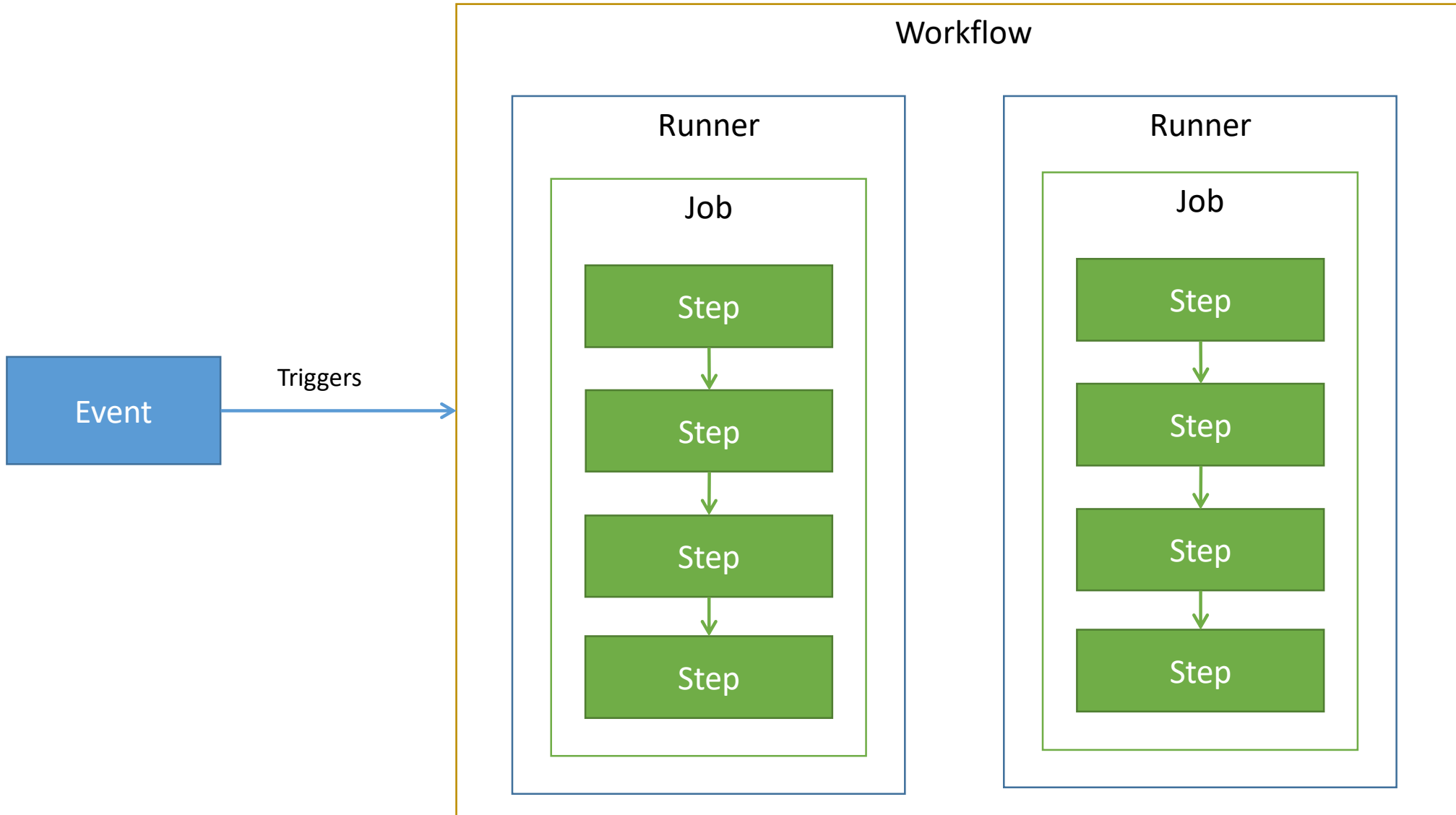
GitHub - Disclaimer



- Microsoft has **Embraced** GitHub and Open Source
- Microsoft is **Extending** GitHub
- Guess what comes next?

- GitHub is a Microsoft's **product**
- Do **not trust** Microsoft
- Do not trust anyone. **Trust the code**
- **Never** depend on a company
- **Be ready** to leave GitHub **now**

GitHub Actions: Overview

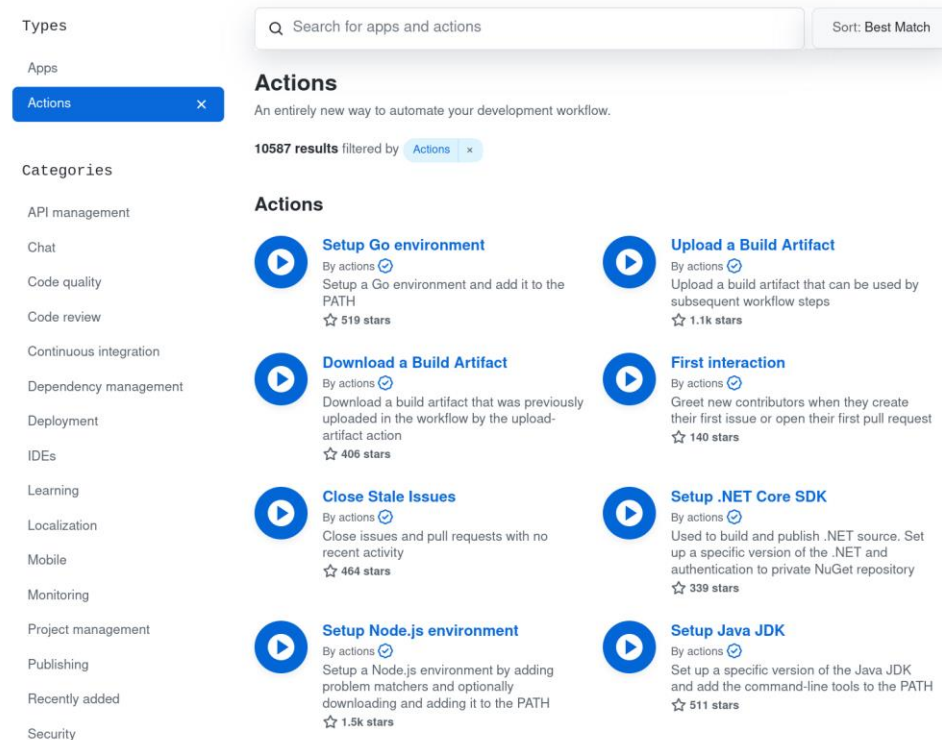


GitHub Actions: Action

A step can be just a sequence of commands or an **Action**

Actions are predefined and community-maintained scripts. Actions can receive inputs and produce outputs. You can write custom actions as Dockerfiles or using Javascript

GitHub marketplace offers a wide repository of community-maintained Actions



The screenshot displays the GitHub Actions marketplace interface. On the left, there is a sidebar with 'Types' (Apps, Actions) and 'Categories' (API management, Chat, Code quality, Code review, Continuous integration, Dependency management, Deployment, IDEs, Learning, Localization, Mobile, Monitoring, Project management, Publishing, Recently added, Security). The 'Actions' type is selected. At the top, there is a search bar and a 'Sort: Best Match' button. Below the search bar, the text 'Actions' is followed by 'An entirely new way to automate your development workflow.' and '10587 results filtered by Actions'. The main area shows a grid of action cards, each with a play button icon, a title, a description, and a star count.

Action Name	Description	Stars
Setup Go environment	Setup a Go environment and add it to the PATH	519 stars
Upload a Build Artifact	Upload a build artifact that can be used by subsequent workflow steps	1.1k stars
Download a Build Artifact	Download a build artifact that was previously uploaded in the workflow by the upload-artifact action	406 stars
First interaction	Greet new contributors when they create their first issue or open their first pull request	140 stars
Close Stale Issues	Close issues and pull requests with no recent activity	464 stars
Setup .NET Core SDK	Used to build and publish .NET source. Set up a specific version of the .NET and authentication to private NuGet repository	339 stars
Setup Node.js environment	Setup a Node.js environment by adding problem matchers and optionally downloading and adding it to the PATH	1.5k stars
Setup Java JDK	Set up a specific version of the Java JDK and add the command-line tools to the PATH	511 stars

Some community Actions

Action	Description
JamesIves/github-pages-deploy-action@4.1.4	Deploy a folder to the gh-pages branch
pypa/gh-action-pypi-publish@master	Publish a package to PyPI
EnricoMi/publish-unit-test-result-action@v1	Generate python's unit test reports
actions/setup-python@v2	Setup a python environment
github/super-linter@v4	Run a combination of linters a generates a report
SamKirkland/FTP-Deploy-Action@4.1.0	Deploy a folder to an FTP server
elgohr/Publish-Docker-Github-Action@master	Deploy an image to the Docker Registry

GitHub Actions: Syntax

GitHub Actions uses YAML to define workflows

```
name: <workflow-name>
on: <events>

jobs:
  my-job-name:
    runs-on: <runner>
    steps:
      - name: <name-of-the-step>
        run: <command-to-execute>
      - name: <name-of-the-step>
        uses: <user>/<action>
        with:
          argument-1: <value-1>
          argument-2: <value-2>
```

Available runners hosted on GitHub:

- ubuntu-<latest | 20.04 | 18.04>
- macos-<latest | 11 | 10.5>
- windows-<latest | 2022 | 2019>

Events can be:

- push
- pull_request
- schedule
- ...

GitHub Actions: Example

```
name: Run unit tests
on: push

jobs:
  run-tests:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout the code
        uses: actions/checkout@main
      - name: Setup python 3.7
        uses: actions/setup-python@v1
        with:
          python-version: 3.7
      - name: Run the tests
        run: python -m unittest
```

GitHub Actions: Containers and Services

You can specify which container to use to run a job.

You can also specify additional containers needed to execute the job.

```
name: Run nodejs tests
on: push

jobs:
  run-tests:
    runs-on: ubuntu-latest
    container: node:lts-alpine
    services:
      redis:
        image: redis
    steps:
      - name: Checkout the code
        uses: actions/checkout@main
      - name: Install Dependencies
        run: npm install
      - name: Run tests that requires redis
        run: npm test
        env:
          REDIS_HOST: redis
```

GitHub Actions: Events

You can specify a list of events to capture. You can specify arguments to each event.

You can even schedule workflows to run using the cron syntax

```
on: [push, pull_request]
```

```
on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main
      - develop
```

```
on:
  schedule:
    - cron: "0 0 * * *"
```

GitHub Actions: Secrets

You can securely store secrets (passwords, tokens, etc...) in GitHub and they will be passed to the jobs. You can access the secrets using `${{ secrets.SECRET_NAME }}`

```
name: Publish to PyPI
on: push

jobs:
  build-n-publish:
    name: Build and publish to PyPI
    runs-on: ubuntu-18.04
    steps:
      - uses: actions/checkout@master
      - name: Set up Python 3.7
        uses: actions/setup-python@v1
        with:
          python-version: 3.7
      - name: Install
        run: python -m pip install build --user
      - name: Build binary wheel and a source tarball
        run: python -m build --sdist --wheel --outdir dist/ .
      - name: Publish to PyPI
        uses: pypa/gh-action-pypi-publish@master
        with:
          password: "${{ secrets.PYPI_API_TOKEN }}"
```

GitHub Actions: Complex Workflows

You can specify dependencies between jobs to create sequential and concurrent workflows

```
name: Run tests and then publish on PyPI
on: push

jobs:
  run-tests:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout the code
        uses: actions/checkout@main
      - name: Set up Python 3.7
        uses: actions/setup-python@v1
        with:
          python-version: 3.7
      - name: Run the tests
        run: python -m unittest
```

```
build-n-publish:
  name: Build and publish to PyPI
  runs-on: ubuntu-18.04
  needs: run-tests
  steps:
    - uses: actions/checkout@master
    - name: Set up Python 3.7
      uses: actions/setup-python@v1
      with:
        python-version: 3.7
    - name: Install
      run: python -m pip install build --user
    - name: Build binary wheel and a source tarball
      run: python -m build --sdist --wheel --outdir dist/ .
    - name: Publish to PyPI
      uses: pypa/gh-action-pypi-publish@master
      with:
        password: ${ secrets.PYPI_API_TOKEN }
```


GitHub Actions: Matrix

You can specify variables ranges and exploration strategies for the same job

```
jobs:
  run-browser-tests:
    runs-on: ubuntu-latest
    strategy:
      matrix:
        node: [6, 8, 10]
        browser: ["firefox", "chrome"]
    steps:
      - name: Setup node-${{ matrix.node }}
        uses: actions/setup-node@v2
        with:
          node-version: ${{ matrix.node }}
      - name: Install dependencies
        run: npm install
      - name: Run tests
        run: npm test
        env:
          BROWSER: ${{ matrix.browser }}
```

GitHub Actions: Code Reuse

Code reusing is a bit messy in GitHub Actions. You have to define a workflow that gets triggered on the event “workflow_call” and use the “uses” keyword in the callee job definition

```
# test.yaml
name: Reusable test
on: workflow_call

jobs:
  run-tests:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout the code
        uses: actions/checkout@main
      - name: Set up Python 3.7
        uses: actions/setup-python@v1
        with:
          python-version: 3.7
      - name: Run the tests
        run: python -m unittest
```

```
# push-pr.yaml
name: Run test on push or PR
on: [push, pull_request]

jobs:
  call-test:
    uses: <user>/<repo>/.github/workflows/test.yaml@main
```

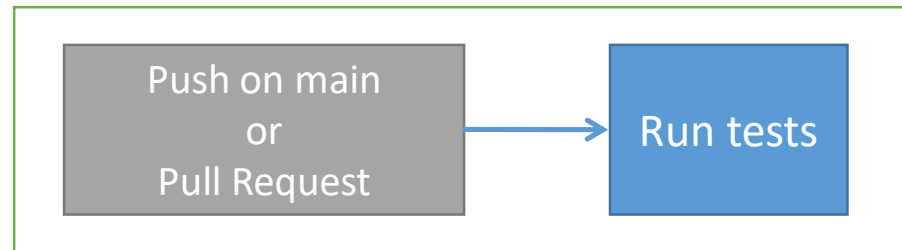
GitHub Actions: Example

We will use as example the development of a python package.

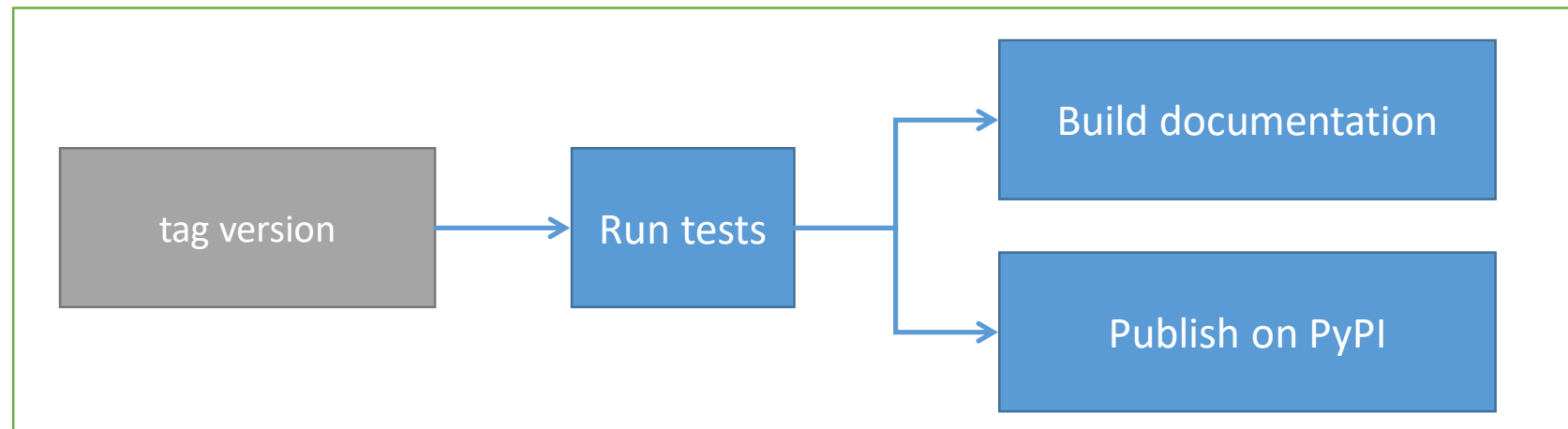
We will define three jobs

- Run the unit test
- Build the documentation and publish it on GitHub Pages
- Build the package and publish it on PyPI

Workflow 1



Workflow 2



Source code on GitHub

<https://github.com/galatolofederico/weirdcipher>